

UNIT 2: CORROSION SCIENCE
REVIEW QUESTIONS FROM PREVIOUS TEST PAPERS

Reasoning questions on corrosion with answers

- 1. With equations, illustrate the formation of rust when tin coated iron with pin holes is exposed to moist atmosphere. (5M)**

Ans) Explanation of area effect with identification of anode and cathode. Cathodic and anodic reactions to be explained.

- 2. Interpret the effect on corrosion rate with respect to temperature and when small anodic area is present compare to large cathode area. (6M)**

Ans) Temperature: As the temperature increases, rate of corrosion increases. Explain with two examples.

Reason: Increase in temperature, increases the conductance of the aqueous medium, hence corrosion progresses faster.

Small anodic area is present compare to large cathode area

A broken coating of tin on iron (tin is cathodic to iron). If some areas of iron is not covered (or) peeled off by tin, it results in smaller anodic area (iron metal) and larger cathodic area (tin coat). In this case, more intense or rapid corrosion is observed at smaller anodic area (iron metal). Iron corrodes faster than tin according to galvanic series.

Reason: As cathode is larger, demand for electrons will be very high and the electrons liberated at the anode are rapidly consumed by the cathode. Hence, an intense or rapid corrosion takes place at the exposed small anodic area. Thus, the rate of corrosion is increased at small anodic area.

- 3. With suitable examples predict the effect of these factors on the rate of corrosion (3M)**
:Relative anodic and cathodic areas

Ans)

ii) Relative anodic and cathodic areas with the following two examples including figures (refer notes).

Example: Tin coated on iron and zinc coated on iron

- 4. Justify the following:**

i) A mild steel tap in copper drum undergoes faster corrosion (or) steel pipe connected to copper plumbing ii) Iron corrodes faster at lower pH (4M)

Ans) i) Differential metal (or) galvanic (or) bimetallic corrosion

Reason: When two dissimilar metals are in contact with each other in corrosive environment, the metal with lower electrode potential becomes anodic and undergoes corrosion, whereas the

metal with higher electrode potential act as cathode and remain unaffected. Steel is an alloy of iron, which is higher in the electrochemical series. So, it forms the anode. Hence, iron undergoes oxidation and gets corroded when steel pipe is connected to copper plumbing is exposed to a corroding environment.

Why does a steel pipe in a large copper tank corrode using causing rapid destruction? (2M)

Ans) Iron or steel is anodic to copper. When steel pipe fitted in a large copper tank is exposed to atmosphere, galvanic corrosion starts and the anodic metal (iron or steel) starts corroding. Moreover, small sized steel pipe (anode) in a large-sized copper tank (cathode) caused rapid and intense corrosion (or destruction) of steel pipe.

ii) Iron corrodes faster at lower pH. In acidic medium, the rate of corrosion depends upon the rate of evolution (liberation) of hydrogen at cathode area.

Example: When pH less than 4, corrosion rate increases due to higher concentration of H^+ ions (due to evolution of hydrogen at the cathodic area). Reaction should be mentioned **(refer notes/materials)**.

5. Explain corrosion control by galvanization process. Why galvanized vessels are not used for food storage. (5M)

Ans) Explanation of galvanization process by hot dipping method.

Galvanized vessels (or) utensils are not used for food storage because Galvanizing process creates a coating to the metal that prohibits rust. This zinc coating which can be toxic when consumed. Cooking utensils and storage containers are commonly not made of galvanized steel.

6. Explain how the following factors affect the rate of corrosion (5M)

i) pH of the medium ii) area of anode to cathode

Ans) i) Different pH of the medium should be explained with suitable reactions.

ii) Factors affecting rate of corrosion-Relative surface area of anodic and cathodic areas
Smaller anodic area (Fe metal) and larger cathodic (tin coating) area with figure and labeling must be represented including the following explanation.

If some areas of iron is not covered (or) peeled off by tin, it results in smaller anodic area (iron metal) and larger cathodic area (tin coat). In this case, more intense or rapid corrosion is observed at smaller anodic area (iron metal). Iron corrodes faster than tin according to galvanic series.

Reason: As cathode is larger, demand for electrons will be very high and the electrons liberated at the anode are rapidly consumed by the cathode. Hence, an intense or rapid corrosion takes place at the exposed small anodic area. Thus, the rate of corrosion is increased at small anodic area.

7. Illustrate corrosion control of an iron object by cathodic protection method (5M)

Ans) Cathodic protection method by taking either sacrificial anodic method with figure, and explanation **(refer notes/materials).**

8. A broken coating of tin over iron deteriorates the iron object. Justify the statement.

(2M)

Ans) A broken coating of tin on iron (tin is cathodic to iron). If some areas of iron is not covered (or) peeled off by tin, it results in smaller anodic area (iron metal) and larger cathodic area (tin coat). In this case, more intense or rapid corrosion is observed at smaller anodic area (iron metal). Iron corrodes faster than tin according to galvanic series.

Reason: As cathode is larger, demand for electrons will be very high and the electrons liberated at the anode are rapidly consumed by the cathode. Hence, an intense or rapid corrosion takes place at the exposed small anodic area. Thus, the rate of corrosion is increased at small anodic area.

9. Justify: i) Phosphating is a surface conversion coating

(1M)

ii) Mg blocks are attached to underground pipelines

(2M)

Ans) i) Phosphating is the process of coating of metal phosphate by oxidizing its outer layer in phosphoric acid (H_3PO_4) medium. It forms a metal phosphate which acts as a protective barrier to the base metal which is undergoing corrosion. Hence, phosphating is a surface conversion coating.

ii) Cathodic protection method by taking either sacrificial anodic method with figure, and explanation **(refer notes/materials).**

10. Explain the type of corrosion taking place when a part of the metal surface is covered under dust particles for a long time

(5M)

Ans) Differential aeration corrosion-Pitting corrosion with Iron as an example, figure, labeling, and reactions at anodic and cathodic regions **(refer notes/materials).**

11. Account for the following:

i) Part of the window grill inside the frame undergoes corrosion (or) a pure metal rod half-immersed vertically in water starts corroding at the bottom (or) ship sailing in water corrodes below water line

(5M)

Ans) Differential aeration corrosion explanation by taking **any one of the following example (refer notes).**

Ex: Iron rod partially immersed in NaCl solution (or) water-line corrosion (or) pitting corrosion. Explain with figure and reactions at anodic and cathodic reactions **(refer notes/materials).**

Reason: When the part of the window grill inside the frame exposed to lower concentration (less oxygenated area), it becomes anodic area and undergoes corrosion. Higher concentration

(more oxygenated area) becomes cathodic area and becomes unaffected. Two different concentration cells are formed.

12. Why does part of a nail inside the wood undergo corrosion easily? (5M)

Ans) The part of nail inside the wood is not exposed to atmospheric condition and becomes deoxygenated thereby behaves as anode. So, due to differential aeration, the part of nail inside the wood acting as anode, undergoes oxidation and gets corroded.

**13. Give reason: i) Higher the hydrogen over voltage lower will the rate of corrosion
ii) The corrosion product formed on aluminium is Al_2O_3 ; it's passive, whereas corrosion product formed on iron is Fe_2O_3 ; it's active (5M)**

Ans) i) A metal with low hydrogen overvoltage is more susceptible (reactive) for corrosion, when the cathodic reaction is hydrogen evolution type. With lower hydrogen overvoltage, hydrogen gas is liberated easily and hence, the cathodic reaction rate is faster.

ii) In case of aluminium, Al_2O_3 is stoichiometric in composition, highly insoluble, stable, uniform, and non-porous with low ionic and electronic conductivities, and then the passive layer prevents further corrosion. On the other hand, Fe_2O_3 is non-stoichiometric, highly soluble, unstable, non-uniform, and porous with high ionic and electronic conductivities, and hence, it's active.

14. Identify and explain the types of corrosion occurring in the following:

i) Iron pipe fixed with a copper tap (or) iron bolts in copper vessel are undesirable

ii) Partially buried iron pipe (6M)

Ans) i) It leads to differential metal corrosion. Iron has low electrode potential which acts as an anode and undergoes corrosion. Copper has high electrode potential which acts as cathode and remains unaffected.

ii) It leads to differential aeration corrosion. Buried part acts as an anode due to less air (poorly oxygenated) and undergoes corrosion. While upper part acts as cathode and it's protected.

15. "Anodic metallic coating is advantageous over cathodic metallic coating (or) cathodic coating is unsafe compared to anodic metal coating. Justify the statement". Outline the process of tinning of Fe article. (6M)

Ans) In case of anodic metallic coatings, anodic coating if breaks or discontinuities occur in such a coating, the base metal is not corroded, till all the coating metal is consumed.

Example: Coating of Zn on iron (Galvanization process)

In case of cathodic metallic coatings, it is unsafe, if it breaks or discontinuities occur in such a coating, the intense corrosion of base metal takes place because of large cathodic area and small anodic area.

Example: Coating of Sn on iron (Tinning process)

In case of anodic metallic coatings, electrode potential of coating metal is lower than that of the base metal. Coating metals are anodic to the base metal.

In case of cathodic metallic coatings, electrode potential of coating metal is higher than that of the base metal. Coating metals are cathodic to the base metal.

Process of tinning

Hot-dipping method: Figure and explanation **(refer notes/materials).**

Galvanization of iron articles is preferred to tinning. Give reason. (2M)

Ans) Zinc is more electropositive than iron, so zinc coating acts as anode; while the exposed iron portions of coating acts as cathode. If by chance, the zinc coating is broken at some place, the zinc (being more anodic than iron), undergoes corrosion, protecting iron (which is cathodic) from rusting. Thus, no attack on the iron occurs, until practically all the zinc has been corroded in the vicinity of the exposed iron spot. So zinc coating protects iron 'sacrificially'.

On the other hand, tin is a noble metal (ie, having higher electrode potential than iron), so it protects the iron due to its higher corrosion resistance than iron. Such a coating provides effective protection to iron only when the coating is completely continuous. If tin coating is punctured, much more corrosion damages can be done to the base metal of iron than to iron metal without it. Since tin becomes cathode; while the exposed iron (which is above tin in the electrochemical series) acts as anode, so a galvanic cell is set up, thereby an intense corrosion at the small exposed iron part occurs.

16. Rusting of iron is quicker in saline water than in ordinary water. Why? (2M)

Ans) Saline water contains common salt, sodium chloride, which leads conductivity of water. When saline water comes in contact with iron, corrosion current increases and rusting of iron takes place rapidly.

17. Why does corrosion of water filled steel tanks occur below the waterline (or) Corrosion of water filled steel tanks occurs below the waterline. Give reason. (2M)

Ans) This is because the area above the waterline is highly-oxygenated and acts as cathodic; while the fully immersed part is poorly-oxygenated and acts as anodic. So the anodic part (below the water line) gets corroded, due to electrochemical corrosion; while the cathodic part (above the waterline) remains completely unaffected by corrosion. This is called differential aeration corrosion.

18. What is galvanic corrosion? (2M)

Ans) When two dissimilar metals are electrically connected and exposed to an electrolyte, the metal higher in electrochemical series undergoes corrosion.

19. Iron corrodes faster than aluminum, even though iron is placed below aluminum in the electrical series. Why? (2M)

Ans) This is due to aluminum forms a non-porous, very thin, tightly adhering protective oxide film (Al_2O_3) on its surface and this film does not permit corrosion to occur.

20. A steel screw in a brass marine hardware corrodes. Give reason. (2M)

Ans) This is due to differential metal (or) galvanic (or) bimetallic corrosion. Iron (higher in series than brass) becomes anodic and attacked; undergoes corrosion. Brass (lower in series) acts as cathodic and it is unaffected.

21. Iron corrodes under drops of salt solution. Give reason. (2M)

Ans) This is due to differential aeration corrosion. Areas of iron covered by drops, having poorly oxygenated areas become anodic. The other areas are freely exposed to air. Due to electrochemical corrosion, the areas under drops (anodic) undergo corrosion; while the freely exposed parts remain unaffected.

Additional notes on only differential metal (or) Galavanic (or) bimetallic corrosion and differential aeration corrosion

Examples for differential metal (or) Galavanic (or) bimetallic corrosion

- 1) Steel screws/rivets in copper sheet
- 2) Steel pipe connected to copper plumbing

Examples for differential aeration corrosion

- 1) Part of the nail inside the wall being exposed to lower oxygen concentration than the exposed part undergoes corrosion
- 2) Window rods inside the frame suffer corrosion but not the exposed regions
- 3) Paper pins inside the paper get corroded and the exposed part is free from corrosion
- 4) Metal under dirt, dust, scale or water under corrosion
- 5) Partially buried pipeline in soil or submerged in water undergoes corrosion below the soil or water where as the exposed part is free from corrosion

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